Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/11/2022
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sdytse@rutgers.edu	**Fax:**	FY 609 258-6233
PI Organization Type:	UNIVERSITY	**Phone:**	609 258-5271
Organization Name:	Princeton University		
PI Address 1:	Department of Mechanical and Aerospace Engineering		
PI Address 2:			
PI Web Page:			
City:	Princeton	**State:**	NJ
Zip Code:	08544	**Congressional District:**	12
Comments:	Chung Law's email is cklaw@princeton.edu		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	11/25/2014	**End Date:**	08/26/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 8/26/2022 per NASA GRC (CoInvestigator Stephen Tse has accompanying grant). (Ed., 11/30/2021) NOTE: End date changed to 8/26/2021 due to CoInvestigator Stephen Tse's accompanying grant that ends 8/26/2021 (Ed., 4/20/20) NOTE: Extended to 11/24/2019 per NSSC info (Ed., 11/25/16)		
Key Personnel Changes/Previous PI:	NOTE April 2020: Stephen Tse, CoInvestigator, continues work under this grant with his grant 80NSSC18M0134, which continues through 8/26/2021; he will continue reporting.		
COI Name (Institution):	Tse, Stephen Ph.D. (Rutgers University grant 80NSSC18M0134)		
Grant/Contract No.:	NNX15AC79A ; 80NSSC18M0134 (CoI Tse)		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of Structure and Response of Spherical Diffusion Flames--NNX09AR03A with the same Principal Investigator, Chung K. Law. NOTE April 2020: Stephen Tse, CoInvestigator, continues work under this grant with his grant 80NSSC18M0134, which continues through 8/26/2021; CoI Tse will continue reporting. The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the Glenn Research Center (GRC) and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
Task Progress:	The s-Flame tests are intended to: (i) Advance our ability to predict the structure and dynamics, including extinction, of both soot-free and sooty flames; (ii) Obtain experimental data in simple, well-defined flow fields and understand various key elemental diffusion flame processes; (iii) Identify the extinction limits for both radiative and convective extinction (i.e., at high and low system Damköhler numbers, respectively); (iv) Determine the existence, onset, and nature of pulsating instabilities that have been theoretically predicted to occur in such flames with fuel/diluent mixtures that are above a critical Lewis number; (v) Validate and/or improve accuracy of computational simulations, including detailed chemistry and transport. The rationale for the matrix of experiments is to utilize various mixtures to assess both chemical and transport effects (including radiative) on flame behavior. Pure H2 and pure CH4 are utilized to assess kinetic mechanisms associated with each. Additionally, pure H2 only produces H2O as its radiative product, thereby isolating its radiative properties. Mixtures composed of both H2 and CH4 are used to examine their combined chemical and transport effects. N2 and He inerts, which affect characteristic flame temperature and transport (molecular and radiative), in the same mole fractions, are utilized for comparison. The project has completed two (2) rounds of microgravity experimentation. Currently, detailed analyses of the acquired data, computational simulation, and theoretical analysis are ongoing. Interpretation of all phenomena will emphasize the simultaneous and coupled influence of chemistry and transport.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Xiong G, Gandhi R, Zhong X, Mädler L, Tse SD. "Binary collision of a burning droplet and a non-burning droplet of xylene: Outcome regimes and flame dynamics." Proceedings of the Combustion Institute. 2019;37:3345-52. https://doi.org/10.1016/j.proci.2018.06.198 , Jan-2019